

Neural Networks for NLP - Sheet 3

25.11.2024

Deadline: 16.12.2024 - 13:15, on OLAT as **pdf/.py/.ipynb** and/or after the lecture

Task 1: Long live Attention

12 Points

- (a) **(2 points)** The *transformer* architecture makes use of two kinds of attention mechanisms (check the transformer architecture in the lecture slides). Briefly talk about the objective of each mechanism with respect to where in the architecture they are used.
- (b) **(6 points)** Given below are some values:

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix}, W_Q = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, W_K = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}, W_V = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 3 & 0 \\ 1 & 0 & 3 \\ 1 & 1 & 0 \end{bmatrix}$$

where X is the input (each row is an input), and W_Q, W_K , and W_V are the weights for query, key, and value respectively. Calculate self-attention for X showing all the intermediate steps (round up your results to 1 or 2 decimal places).

- (c) **(4 points)** Perform the same steps as in part (b) in the notebook provided with this exercise sheet using simple Pytorch operations, and verify that you get the same results as in part (b).

Task 2: MyGPT: Building a GPT-style language model

10 Points

In this task, your objective is to implement the key components of a Generative Pretrained Transformer (GPT) language model using PyTorch. To this end, you will use the `myGPT.ipynb` notebook as the skeleton for your code, where you will find step-by-step instructions to implement

- (a) **(2 points)** the self-attention head
- (b) **(2 points)** the multi-head self-attention module
- (c) **(2 points)** the feed forward layer
- (d) **(2 points)** the transformer block
- (e) **(2 points)** the myGPT model using the components above

For each of the above components, fill in the code following the instructions, and explain your code using comments.

Task 3: Generating Text from Language Models

5 Points

Inside the Materials folder, there's a file named `tutorial3.txt`, which contains the entire collection of William Shakespeare's works. Your objective is to develop a language model capable of producing text in the style of Shakespeare. To this end, you are going to train the GPT model you built in Task 2. Here's what you need to do:

- (a) **(2 points)** Construct the training set for your model and run the provided code to train it.
- (b) **(3 points)** Extend the `myGPT` model with a text generation function using the greedy decoding strategy, which selects the token with the highest probability at each successive step. Use this function to create a 1000-character long text starting with the phrase "To be or not to be," and briefly comment on the quality of the generated passage.

Use the notebook `myGPT.ipynb` as your base code.

Task 4: Better decoding strategies

9 Points

In text generation, greedy decoding often leads to repetitive and less diverse outputs, as it always chooses the most probable next token without considering the overall sequence quality or long-term context. This exercise aims to explore the efficacy of other decoding strategies. Based on the language model in Task 3:

- (a) **(3 points)** Use the top-K sampling method to create a 1000-character long text, starting with the phrase "To be or not to be, ". Try different values of K .
- (b) **(3 points)** Temperature T is a hyperparameter that can be used to control the randomness of predictions. This is done by dividing the logits by T prior to the application of the softmax function. Improve the previous implementation to include temperature T and regenerate the 1000-character text. Try different temperature values. How does it affect the generated text?
- (c) **(3 points)** Use the beam search method to create a 500-character long text. This passage should commence with the phrase "To be or not to be,". A critical aspect of this task is the experimental variation of the `beam_width` parameter and to report the impact of different beam width values on the generated text's coherence, stylistic alignment with Shakespearean prose, and overall linguistic quality.

Use the notebook `myGPT.ipynb` as your base code for this task.

Important Notes:

1. You need to complete and run the entire code in the notebook given to you as part of this assignment. Submitting the completed notebook containing your results is **mandatory**. For theoretical answers, you may add them in the notebook (in text cells) or you can also submit your answers to such questions in a separate PDF file. Remember, submission of code in PDF file is **strictly not allowed**.
2. Any kind of plagiarism and cheating (from your classmates as well as from the internet) is **strictly forbidden** and may lead to your disqualification from the course. You have been warned!